

Global Population and Demographic Data Analysis Using Machine Learning

1. Introduction

For this project, Global Population and Demographic Data was collected through web scraping from Worldometer. The collected dataset contains population-related information for different countries and different years.

After collecting the data, a data cleaning process was performed to remove duplicate records, handle missing values, convert numerical values into appropriate data types, and detect invalid values.

The cleaned dataset contains the following important features:

- Country
- Year
- Population
- Yearly_Percent_Change
- Yearly_Change
- Migrants_Net
- Median_Age
- Fertility_Rate
- Density_Per_Km2
- Urban_Population_Percent
- Urban_Population
- Country_Share_of_World_Population
- World_Population
- Global_Rank

Using this dataset, several Data Science and Machine Learning projects can be implemented. This proposal report presents four different proposals based on the collected and cleaned dataset.

Proposal 1: Population Prediction Using Regression

1.1 Proposed Work

The first proposed project is to predict the future population of a country using historical population data.

Population changes over time. Therefore, historical information such as Year, previous Population, Yearly Change, Fertility Rate, Migration, Median Age, and Urban Population can be used to predict population values.

The main objective of this project is to develop a machine learning model that can estimate the population of a country for a particular year.

1.2 Input Features

The following features can be used as input:

- Year
- Yearly_Percent_Change
- Yearly_Change
- Migrants_Net
- Median_Age
- Fertility_Rate
- Density_Per_Km2
- Urban_Population_Percent
- Urban_Population

The target variable will be: Population

1.3 Proposed Models

The following Regression models can be implemented:

- Linear Regression
- Random Forest Regressor
- Decision Tree Regressor
- Gradient Boosting Regressor

Among these models, Random Forest Regression can be useful because it can identify complex relationships between multiple demographic variables.

1.4 Methodology

The dataset will first be divided into training and testing datasets.

For example:

- Training Data: 80%
- Testing Data: 20%

The selected machine learning model will be trained using the training dataset.

After training, the model will predict the population values for the testing dataset.

The predicted population values will be compared with the actual population values.

1.5 Expected Result

The system will produce:

- Predicted Population
- Actual Population
- Prediction Error
- Population Growth Trend

The model performance can be evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

1.6 Expected Output

Example Output:

Country: Bangladesh

Year: 2030

Predicted Population: XXXXXXXX

Actual Population: XXXXXXXX

The final result can help researchers understand future population growth trends.

Proposal 2: Population Growth Classification

2.1 Proposed Work

The second proposed project is to classify countries based on their population growth condition.

Countries can experience different types of population growth. Some countries may have high population growth, while others may have moderate or low population growth.

Using the Yearly_Percent_Change feature, population growth can be categorized into different classes.

For example:

- Low Growth
- Medium Growth
- High Growth

A new target column called Population_Growth_Category can be created.

2.2 Data Preparation

The Yearly_Percent_Change feature can be used to create categories.

For example:

Low_Growth:

Population growth less than a selected threshold.

Medium_Growth:

Population growth within a moderate range.

High_Growth:

Population growth above a selected threshold.

The threshold can be determined by analyzing the dataset distribution.

2.3 Input Features

The following features can be used:

- Fertility_Rate
- Median_Age
- Migrants_Net
- Density_Per_Km2
- Urban_Population_Percent
- Yearly_Change

The target variable will be: Population_Growth_Category

2.4 Proposed Models

The following Classification models can be implemented:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors (KNN)

Random Forest Classification can be useful because multiple demographic features may influence population growth.

2.5 Methodology

The dataset will be divided into training and testing data.

The machine learning model will learn the relationship between demographic factors and population growth categories.

After training, the model will classify new records into:

- Low Growth
- Medium Growth
- High Growth

2.6 Expected Result

The system will identify which demographic characteristics are related to high or low population growth.

For example:

High Fertility Rate + High Yearly Change → High Growth

Low Fertility Rate + High Median Age → Low Growth

2.7 Expected Output

Example Output:

Country: Afghanistan

Predicted Growth Category: High Growth

Confidence Score: 92%

The result can help analyze the population growth pattern of different countries.

Proposal 3: Country Clustering Based on Demographic Characteristics

3.1 Proposed Work

The third proposed project is to group countries based on similar demographic characteristics.

This project will use Unsupervised Machine Learning.

Countries with similar population characteristics will be placed in the same group.

For example, countries can be grouped based on:

- Population
- Fertility Rate
- Median Age
- Population Density
- Urban Population Percentage
- Migration

3.2 Input Features

The following features can be used:

- Population
- Fertility_Rate
- Median_Age
- Density_Per_Km2
- Urban_Population_Percent
- Migrants_Net

Before implementing the model, the numerical features will be normalized using feature scaling.

3.3 Proposed Model

The main model will be: K-Means Clustering

Other clustering algorithms may also be tested.

The optimal number of clusters can be determined using:

- Elbow Method
- Silhouette Score

3.4 Methodology

First, demographic features will be selected.

Then the data will be standardized.

After that, the K-Means algorithm will divide countries into several clusters.

For example:

Cluster1:

High Population and High Density Countries

Cluster2:

Low Population and Low Density Countries

Cluster3:

High Fertility Rate Countries

Cluster4:

Highly Urbanized Countries

3.5 Expected Result

The model will automatically discover similarities between countries.

Countries with similar demographic characteristics will belong to the same cluster.

This analysis can help identify different demographic patterns around the world.

3.6 Expected Output

Example Output:

Cluster1:

Countries with high population and high population density.

Cluster2:

Countries with low fertility rates and high median age.

Cluster3:

Countries with high fertility rates and rapid population growth.

The results can be visualized using:

- Scatter Plot
- Cluster Visualization
- PCA Visualization

Proposal 4: Population Trend Forecasting

4.1 Proposed Work

The fourth proposed project is to analyze historical population trends and forecast future population growth.

Since the dataset contains population data from different years, time-based analysis can be performed.

The objective is to identify population growth patterns and estimate future population values.

4.2 Input Data

The primary features will be:

- Country
- Year
- Population

Additional demographic features can also be included:

- Fertility_Rate
- Migrants_Net
- Median_Age
- Urban_Population_Percent

4.3 Proposed Models

The following forecasting models can be implemented:

- Linear Regression
- Polynomial Regression
- ARIMA
- Random Forest Regressor

For basic implementation, Linear Regression can be used.

For advanced time-series forecasting, ARIMA can be implemented.

4.4 Methodology

First, the dataset will be sorted based on:

Country → Year

Then historical population data will be analyzed.

The model will learn the relationship between Year and Population.

After training, the model can predict future population.

For example:

Historical Data:

2010 → Population

2015 → Population

2020 → Population

2025 → Population

Future Prediction:

2030 → Predicted Population

2035 → Predicted Population

2040 → Predicted Population

4.5 Expected Result

The model will generate:

- Historical Population Trend
- Future Population Prediction
- Population Growth Rate
- Comparison Between Actual and Predicted Population

4.6 Expected Output

Example Output:

Country: Bangladesh

Year: 2030

Predicted Population: XXXXXXXXX

Year: 2035

Predicted Population: XXXXXXXXX

The result can be displayed using a line graph.

The graph will show:

Historical Population → Actual Data

Future Population → Predicted Data

Comparison of the Four Proposals

Proposal	Project Type	Model	Main Output
Proposal 1	Regression	Linear Regression, Random Forest	Predicted Population
Proposal 2	Classification	Random Forest, Decision Tree, KNN	Population Growth Category
Proposal 3	Clustering	K-Means Clustering	Groups of Similar Countries
Proposal 4	Time Series Forecasting	ARIMA, Linear Regression	Future Population Forecast